

SHADOW REVENUE DETECTION SYSTEM

Enterprise-Grade Revenue Leakage Detection using PySpark, Spark SQL, and Medallion Architecture

PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the Successful Completion of the Data Engineering Internship

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Project Technologies

- PySpark
- Spark SQL
- Databricks
- Medallion Architecture

- Delta Lake
- Parquet
- Python
- Data Engineerin

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INTRODUCTION

1.1 Introduction

In today's digital era, retail and e-commerce organizations generate enormous volumes of transactional data every day through online purchases, payment gateways, product catalogs, and customer interactions. This data plays a crucial role in financial reporting, business analytics, and strategic decision-making. However, due to the involvement of multiple systems and continuous data generation, inconsistencies often arise between orders, payments, and product information. These inconsistencies can lead to hidden financial losses that remain unnoticed in conventional reporting systems.

One of the major challenges faced by organizations is the presence of **shadow revenue**, which refers to revenue that is expected but is either lost, inaccurately recorded, or cannot be properly reconciled because of data quality issues. Problems such as missing payments, orphan payments, duplicate order records, incorrect product pricing, and delayed transaction updates significantly affect revenue accuracy and business performance. If these issues are not identified and resolved in time, they may lead to incorrect financial reports, reduced profitability, and poor business decisions.

The **Shadow Revenue Detection System** is designed to address these challenges by building a robust end-to-end data engineering pipeline capable of processing large-scale retail transaction data. The project utilizes **PySpark** and **Spark SQL** within the **Databricks** environment to efficiently process and analyze large datasets. The pipeline follows the **Medallion Architecture**, which organizes data into three progressive layers—**Bronze**, **Silver**, and **Gold**—to improve data quality, maintain consistency, and generate reliable analytical outputs.

The **Bronze Layer** is responsible for ingesting raw data from multiple sources, including Orders, Payments, Products, and Customers, while preserving the original data without modification. The **Silver Layer** focuses on improving data quality by performing operations such as duplicate removal, null value handling, data type conversion, date standardization, and product history management using **Slowly Changing Dimension (SCD) Type 2**. Finally, the **Gold Layer** performs advanced analytics by joining the cleaned datasets, detecting revenue-related anomalies, calculating business Key Performance Indicators (KPIs), and producing a comprehensive **Revenue Integrity Report** for business stakeholders.

A unique feature of this project is the comparison between a **naive (Bad) pipeline** and an **optimized (Good) pipeline**. The bad pipeline processes raw data without applying proper data quality techniques, whereas the good pipeline performs complete cleansing, validation, and transformation before analysis. This comparison clearly demonstrates how effective data engineering practices improve revenue accuracy, eliminate duplicate records, reduce financial discrepancies, and provide more reliable business insights.

The system identifies several critical revenue anomalies, including **Missing Payments**, **Orphan Payments**, **Duplicate Orders**, **Price Mismatches**, and **Date/Time Drift**. After detecting these inconsistencies, the pipeline computes important business metrics such as **Total Revenue**, **Total Payments**, **Revenue Difference**, **Accuracy Ratio**, **Missing Payments Count**, **Orphan Payments Count**, and **Price Mismatch Count**. These metrics provide organizations with a clear understanding of their financial health and help in identifying potential sources of revenue leakage.

The final outcome of the project is an interactive **Revenue Integrity Dashboard**, which presents KPIs, revenue comparisons, anomaly distributions, revenue trends, and leakage analysis through various visualizations. These dashboards enable business users and management teams to monitor revenue integrity effectively and make informed decisions based on accurate, high-quality data.

Overall, the Shadow Revenue Detection System demonstrates how modern data engineering technologies and scalable data processing frameworks can transform raw and inconsistent transactional data into meaningful business intelligence. By combining efficient data processing, systematic data quality management, and advanced analytical techniques, the project provides a practical solution for detecting hidden financial leakages and improving the accuracy of revenue reporting in modern retail environments.

PROBLEM STATEMENT

Modern retail and e-commerce organizations process thousands of transactions every day across multiple systems such as order management, payment gateways, and product catalogs. Since these systems operate independently, inconsistencies often arise in transactional data, making it difficult to maintain accurate financial records.

Common issues include missing payments, orphan payments, duplicate order records, and price mismatches between transaction data and the product catalog. These anomalies lead to hidden revenue leakages, inaccurate financial reporting, and poor business decision-making. Detecting such discrepancies manually is time-consuming and impractical when dealing with large volumes of data.

The **Shadow Revenue Detection System** addresses these challenges by developing a scalable data engineering pipeline using **PySpark**, **Spark SQL**, and the **Medallion Architecture**. The system cleans and transforms raw transactional data, detects revenue-related anomalies, calculates key business metrics, and generates a **Revenue Integrity Report**. This enables organizations to identify hidden financial losses, improve data quality, and support accurate, data-driven business decisions.

OBJECTIVES

The primary objective of the **Shadow Revenue Detection System** is to develop an end-to-end data engineering pipeline that identifies hidden revenue leakages in retail transaction data and improves the accuracy of financial reporting.

The specific objectives of this project are:

- To design and implement a scalable data engineering pipeline using **PySpark** and **Spark SQL**.
- To ingest and integrate data from multiple sources, including Orders, Payments, Products, and Customers.
- To clean and transform raw data by removing duplicates, handling missing values, and standardizing data formats.
- To detect revenue-related anomalies such as missing payments, orphan payments, duplicate orders, and price mismatches.
- To calculate key business metrics, including Total Revenue, Total Payments, Revenue Difference, and Accuracy Ratio.
- To compare the results of a naive (Bad) pipeline with an optimized (Good) pipeline to demonstrate the importance of data quality.
- To generate a **Revenue Integrity Report** and dashboard that provide meaningful insights for business decision-making.

TOOLS & TECHNOLOGIES USED

The development of the **Shadow Revenue Detection System** involved the use of modern data engineering tools and technologies for data processing, transformation, storage, and analytics. The following tools were used during the implementation of the project:

Tool / Technology	Purpose
Python	Used for generating synthetic datasets and implementing data processing logic.
PySpark	Used for distributed data processing, data cleansing, transformations, and anomaly detection.
Spark SQL	Used to perform SQL-based queries, joins, aggregations, and KPI calculations.
Databricks	Cloud-based platform used to develop, execute, and manage the PySpark pipeline.

Tool / Technology	Purpose
Medallion Architecture	Organized data into Bronze, Silver, and Gold layers for structured data processing.
Parquet	Columnar storage format used for efficient data storage and retrieval.
Delta Tables	Used to store processed data and KPI tables with improved reliability and performance.
Git & GitHub	Used for version control and project repository management.
Microsoft Word	Used for preparing the project documentation and final report.

. SYSTEM ARCHITECTURE

The **Shadow Revenue Detection System** follows the **Medallion Architecture**, which organizes data into three layers: **Bronze, Silver, and Gold**. This layered approach improves data quality, simplifies processing, and ensures reliable analytics.

1. Bronze Layer (Raw Data Ingestion)

The Bronze layer is responsible for collecting raw data from different sources, including **Orders, Payments, Products, and Customers**. The data is stored in its original format as Parquet files without any modifications, preserving the source information for future processing.

2. Silver Layer (Data Cleansing and Transformation)

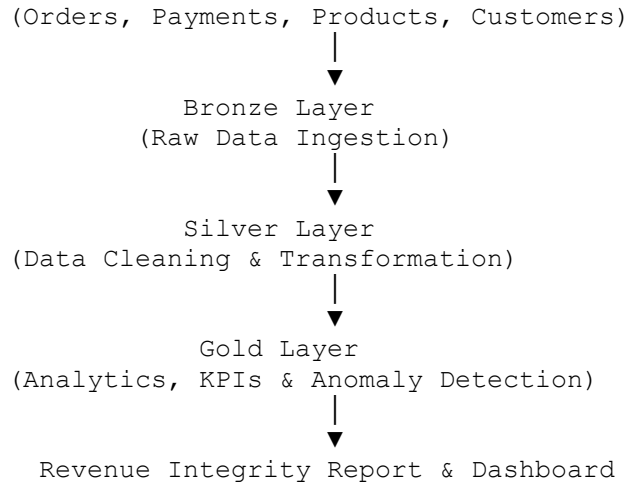
The Silver layer focuses on improving data quality. Operations such as duplicate removal, null value handling, data type conversion, date formatting, and SCD Type 2 processing are performed using PySpark. This layer produces clean and standardized datasets ready for analysis.

3. Gold Layer (Analytics and Reporting)

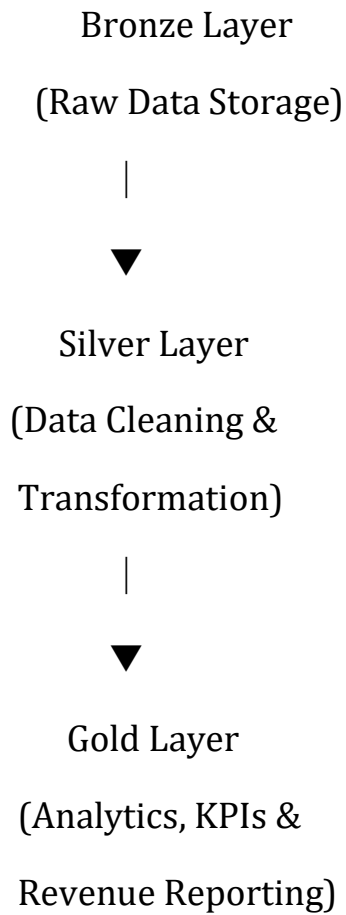
The Gold layer performs advanced data analysis by joining the cleaned datasets to detect anomalies such as missing payments, orphan payments, duplicate orders, and price mismatches. It also calculates key performance indicators (KPIs) like Total Revenue, Total Payments, Revenue Difference, and Accuracy Ratio. The processed data is used to generate the final **Revenue Integrity Report** and dashboard for business stakeholders.

System Architecture Flow

Source Data



Medallion Architecture



Project Workflow

Generate Data



Data Ingestion



Data Cleaning



Transformation



Anomaly Detection



KPI Calculation



Dashboard

DATASET DESCRIPTION

The **Shadow Revenue Detection System** uses four primary datasets representing different components of a retail business. These datasets are processed through the Medallion Architecture to detect revenue-related anomalies and generate meaningful business insights.

Dataset	Description	Key Attributes
Orders	Contains information about customer orders placed in the system. It may include duplicate records and incorrect pricing.	Order ID, Product ID, Customer ID, Quantity, Price, Order Date, Order Status
Payments	Stores payment details for customer orders. It is used to identify missing and orphan payments.	Order ID, Payment Amount, Payment Method, Payment Date
Products	Maintains product catalog information with historical pricing using Slowly Changing Dimension (SCD) Type 2 .	Product ID, Product Name, Price, Effective Date, End Date, Current Flag
Customers	Contains customer information used to associate transactions with individual customers.	Customer ID, Customer Name, City, Signup Date

These datasets are integrated and processed using **PySpark** and **Spark SQL** to clean the data, identify anomalies, calculate business KPIs, and generate the final **Revenue Integrity Report**.

METHODOLOGY

The **Shadow Revenue Detection System** follows a structured methodology based on the **Medallion Architecture**, which organizes data into three processing layers: Bronze, Silver, and Gold. This approach ensures that raw data is progressively transformed into accurate and meaningful business insights.

Step 1: Data Ingestion (Bronze Layer)

Raw datasets containing **Orders, Payments, Products, and Customers** are ingested into the system and stored as Parquet files. This layer preserves the original data without any modifications.

Step 2: Data Cleaning (Silver Layer)

The raw data is cleaned and standardized using **PySpark**. Duplicate records are removed, missing values are handled, data types are converted, and product information is processed using **SCD Type 2**. This ensures high-quality and consistent data for analysis.

Step 3: Data Transformation and Analysis (Gold Layer)

The cleaned datasets are joined using **Spark SQL** to calculate revenue and detect anomalies such as missing payments, orphan payments, duplicate orders, and price mismatches. Business KPIs including Total Revenue, Total Payments, Revenue Difference, and Accuracy Ratio are also computed.

Step 4: Reporting and Visualization

The final processed data is used to generate a **Revenue Integrity Report** and dashboard. The dashboard presents key metrics and visualizations that help stakeholders identify revenue leakages and make informed business decisions.

IMPLEMENTATION

The implementation of the **Shadow Revenue Detection System** was carried out using **PySpark, Spark SQL, and Databricks**. The project follows the Medallion Architecture to process retail transaction data efficiently and identify hidden revenue leakages.

The implementation begins with the ingestion of raw datasets, including **Orders, Payments, Products, and Customers**, into the **Bronze Layer**. These datasets are stored in Parquet format to preserve the original data for further processing.

In the **Silver Layer**, PySpark is used to clean and transform the data. Duplicate records are removed using window functions, missing values are handled, data types are converted to appropriate formats, and product information is processed using **Slowly Changing Dimension (SCD) Type 2**. These operations improve data quality and ensure consistency across all datasets.

The cleaned data is then processed in the **Gold Layer**, where Spark SQL is used to join multiple datasets and detect revenue-related anomalies. The system identifies missing payments, orphan payments, duplicate orders, and price mismatches. It also calculates

important business KPIs such as **Total Revenue**, **Total Payments**, **Revenue Difference**, and **Accuracy Ratio**.

Finally, the processed data is stored as Gold tables and used to generate the **Revenue Integrity Report** and dashboard. The dashboard provides visual insights into revenue performance and highlights areas of financial leakage, enabling stakeholders to make accurate and informed business decisions.

CONCLUSION

The **Shadow Revenue Detection System** demonstrates how modern data engineering techniques can be used to identify and analyze hidden revenue leakages in retail transaction data. By leveraging **PySpark**, **Spark SQL**, and the **Medallion Architecture**, the project successfully transforms raw and inconsistent data into accurate and reliable business information.

The system effectively detects anomalies such as **missing payments**, **orphan payments**, **duplicate orders**, and **price mismatches** while calculating important business KPIs, including Total Revenue, Total Payments, Revenue Difference, and Accuracy Ratio. The comparison between the **Bad** and **Good** pipelines highlights the importance of data cleaning and quality management in producing accurate financial reports.

Overall, the project provides a scalable and efficient solution for revenue integrity analysis. It enables organizations to improve data quality, reduce financial discrepancies, and support better business decision-making through reliable reporting and insightful dashboards.